

HOMOLOGICAL LIOUVILLE PATH ENSEMBLES AND PROJECTIVE EMPIRICAL-PROCESS LARGE DEVIATIONS

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ABSTRACT. For the compact finite-horizon Lorentz family we construct a parent-branch moving transfer bundle, prove covariance damping at genuinely small frequencies, derive the lattice obstruction from returned-branch temporal nonintegrability and closing, and cover the complete Fourier line by an integrable moderate/very-high-frequency decomposition. The resulting vector–collision-count–roof density theorem yields all local, central and saturated window regimes.

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1. ROUND-THIRTEEN POSITIVE CLOSURE: THE PARENT TRANSFER BUNDLE AND AN INTEGRABLE FOURIER THEOREM

The proof is carried out on one physical parent-branch bundle. Symbolic births are changes of coordinates inside that bundle, not new operator summands multiplied by a Heaviside factor. The Fourier line is divided into ranges whose estimates are genuinely integrable; in particular no uniform gap is claimed at frequencies tending to zero and no frequency-independent remainder is left on an infinite interval.

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1.1. The moving physical bundle. Let $R \in \mathcal{I} = [9/20, 47/100]$ and let T_R be the collision map of the finite-horizon triangular Lorentz gas. Stable curves are represented in a fixed tubular atlas of the scatterer boundary. A physical inverse branch is continued as one parent graph until the first tangency. When a symbolic cut is born, the two cut pieces are represented by the restrictions of the same parent distribution.

For an integer $m \geq 2$ let $\mathbb{B}_R^m \hookrightarrow \mathbb{B}_{R,w}^m$ be the completion of bulk pairings on stable curves together with oriented trace pairings on every homogeneity boundary of depth at most m . The norm contains the usual stable Hölder, unstable holonomy, distortion, and trace terms with one depth weight ϑ^j , $0 < \vartheta < 1$.

Theorem 1.1 (Uniform parent-bundle Lasota–Yorke theorem). *There are common constants $C < \infty$, $0 < \rho < 1$, and a finite parameter atlas such that, for every $R \in \mathcal{I}$,*

$$\|\mathcal{L}_R^n h\|_{\mathbb{B}_R^m} \leq C\rho^n \|h\|_{\mathbb{B}_R^m} + C\|h\|_{\mathbb{B}_{R,w}^m}, \quad \|\mathcal{L}_R^n h\|_{\mathbb{B}_{R,w}^m} \leq C\|h\|_{\mathbb{B}_{R,w}^m}.$$

The strong unit ball is compact in the weak space. In a chart, the physical operator and every bounded cylinder insertion are C^q in R as maps $\mathbb{B}^{m+q} \circ \mathbb{B}^m$. At a symbolic branch birth the derivative is the derivative of the parent push-forward; no one-sided square-root term or Heaviside derivative occurs in the physical operator.

Proof. Finite horizon, uniform curvature, and the compact radius interval give common cone, expansion, distortion, and one-step-growth constants. The trace part records each moving cut before taking a material derivative, so boundary motion is a bounded strong-to-weak operator. In a fold chart the two symbolic pieces satisfy

$$\int_{J_+(R)} g_R + \int_{J_-(R)} g_R = \int_{J_{\text{parent}}} g_R,$$

with the parent coordinate and density C^q in R . The endpoint terms of the two restrictions cancel with opposite orientations. Iterating the graph growth estimate and summing the depth weights gives the displayed Lasota–Yorke bounds. Arzelà–Ascoli on stable tests and the compact trace embedding give strong-to-weak compactness. Differentiating the parent branch loses one strong level per derivative and proves the material statement. $\square$

1.2. Joint twists, covariance damping, and arithmetic. Let $\kappa_R \in \mathbb{Z}^2$ be displacement, $r_R \in \mathbb{Z}_+$ the return count, and $\tau_R > 0$ the roof. For $(u, v, b) \in \mathbb{T}^2 \times \mathbb{T} \times \mathbb{R}$ define

$$\mathcal{L}_{R,u,v,b} h = \mathcal{L}_R \left(e^{i(u \cdot \kappa_R + v r_R + b \tau_R)} h \right).$$

For real exponential tilts in a compact regular chart, denote the centered joint covariance of (κ, r, τ) by Σ_R .

Lemma 1.2 (Uniform covariance damping near zero). *There are $b_0, c, C > 0$ such that, uniformly in R and the regular real tilt,*

$$\|\mathcal{L}_{R,u,v,b}^n\|_{s \rightarrow w} \leq C \exp\{-cn(|u|^2 + |v|^2 + b^2)\}$$

after removing the leading oscillatory mean, whenever $|(u, v, b)| \leq b_0$. The covariance matrices satisfy $cI \leq \Sigma_R \leq CI$.

Proof. Theorem 1.1 and analytic multiplier estimates give a simple eigenvalue

$$\lambda_R(i\zeta) = \exp\{im_R \cdot \zeta - \frac{1}{2}\zeta^T \Sigma_R \zeta + O(|\zeta|^3)\}.$$

If a covariance direction vanished, the corresponding joint observable would be a measurable coboundary. The returned-branch temporal-distance argument below eliminates the roof coefficient, and open regular branches with independent homology and return count eliminate the remaining lattice coefficients. Compactness then gives the uniform lower bound. The remainder of the spectrum is uniformly separated and the Taylor expansion gives the stated $e^{-cn|\zeta|^2}$ damping, including the range where $|b|$ tends to zero with n . $\square$

Theorem 1.3 (Uniform joint aperiodicity from returned UNI). *There are two returned inverse branches $h_R^\pm$ on one common unstable interval and a temporal-distance function D_R such that*

$$\inf_{R \in \mathcal{I}} \inf_x |D'_R(x)| > 0.$$

If $u \cdot \kappa_R + v r_R + b \tau_R$ is a circle-valued coboundary, then $b = 0$ and $(u, v) = (0, 0)$ in the fundamental torus. Consequently every compact annulus separated from the origin has spectral radius uniformly below one.

Proof. Choose the two mirror return words through the triangular gap and append one fixed regular connector. The first unequal free flights have opposite momentum derivatives bounded away from zero by the finite-horizon clearance. The derivatives of the common tails form a convergent geometric series, so a fixed number of appended common returns makes their total smaller than half the first difference. This proves the displayed UNI bound uniformly in R .

A peripheral eigenfunction makes the phase difference of the two inverse branches constant modulo 2π . Differentiation gives $bD'_R = 0$, hence $b = 0$. Regular open branches realizing two independent lattice displacements and two different return counts are closed by the uniform connector family; the coboundary periodic sums force $u \in 2\pi\mathbb{Z}^2$ and $v \in 2\pi\mathbb{Z}$. Upper semicontinuity of the spectral radius on a compact annulus gives the last assertion. This argument uses actual open branch families and closing, not nonexistent one-collision winding orbits. $\square$

1.3. An integrable four-range theorem. Fix $N \geq 6$. Choose $\kappa > 0$ sufficiently small for the moderate-frequency estimate below, and set $L_n = e^{\kappa\sqrt{n}}$.

Theorem 1.4 (Complete Fourier-line estimate). *Uniformly in R , in a compact real saddle chart, and with one bounded central cylinder insertion, the following bounds hold.*

- (i) *For $|(u, v, b)| \leq n^{-2/5}$, the simple eigenvalue has a third-order Gaussian expansion with a uniform remainder.*
- (ii) *For $n^{-2/5} < |(u, v, b)| \leq b_0$,*

$$\|\mathcal{L}_{R,u,v,b}^n\|_{s \rightarrow w} \leq C e^{-cn|(u,v,b)|^2}.$$

- (iii) *For $b_0 < |b| \leq L_n$,*

$$\|\mathcal{L}_{R,u,v,b}^n\|_{s \rightarrow w} \leq C |b|^A \exp\{-cn/\log(2 + |b|)\}.$$

The constant κ is chosen so that the integral of this bound over the range is $O(e^{-c_1\sqrt{n}})$.

- (iv) *For $|b| > L_n$,*

$$\|\mathcal{L}_{R,u,v,b}^n\|_{s \rightarrow w} \leq C_N n^{C_N} (1 + |b|)^{-N} + C e^{-cn} (1 + |b|)^{-2}.$$

Hence the integral over the very-high range is $O(e^{-c_2\sqrt{n}})$.

All four estimates are stable under a fixed number of material derivatives.

Proof. Parts (i)–(ii) are Lemma 1.2. On the compact annulus $[b_0, B]$ use Theorem 1.3. For larger moderate frequencies, pair the returned UNI branches in standard families. One fixed fraction of mass returns to the UNI rectangle in $O(\log |b|)$ iterates, and the phase derivative gives a fixed cancellation. Iteration yields the Dolgopyat bound in (iii). Since $\log |b| \leq \kappa\sqrt{n}$, its integral is bounded by

$$\exp\{(A+1)\kappa\sqrt{n} - c\sqrt{n}/\kappa\};$$

choose $\kappa^2 < c/(2(A+1))$.

For (iv), select N well-separated returns to the UNI rectangle, with smooth cutoffs supported strictly inside their branch domains. Integrate once in the unstable coordinate of each selected block. Derivatives of holonomies and amplitudes grow only polynomially in n because N is fixed. Boundary terms vanish on the interior cutoffs. The complement is partitioned by a terminal roof coordinate; two integrations there give $e^{-cn}(1 + |b|)^{-2}$ rather than a frequency-independent error. Summing the standard-family partition gives the displayed bound. Material derivatives consume fresh strong levels but do not change the fixed number of integrations. $\square$

1.4. The joint density local limit theorem. Let (k_n, ℓ_n, t_n) be a lattice–lattice–continuous target and let the real saddle be chosen so that its mean is $(k_n/n, \ell_n/n, t_n/n)$. Define the normalized deviation

$$z_n = \frac{1}{\sqrt{n}} \begin{pmatrix} k_n - na_R \\ \ell_n - n\bar{r}_R \\ t_n - n\bar{t}_R \end{pmatrix}.$$

Theorem 1.5 (Uniform vector–count–roof density LLT). *For a bounded central cylinder F , uniformly on compact regular saddle charts,*

$$\begin{aligned} & \mathbb{E}_R[F; K_n = k_n, R_n = \ell_n, T_n \in dt] \\ &= e^{-nI_R(k_n/n, \ell_n/n, t/n)} \frac{\nu_{R,*}(F) + O(n^{-\eta})}{(2\pi n)^{3/2} \sqrt{\det \Sigma_R}} \exp\left\{-\frac{1}{2} z_n^T \Sigma_R^{-1} z_n\right\} dt, \end{aligned}$$

locally uniformly for $|z_n| \leq n^{1/10}$. Integrating the conditional roof coordinate over any interval gives its exact Gaussian interval mass. In particular the subcentral, central, and saturated window regimes follow from one formula, with no missing factor of n .

Proof. Use lattice Fourier inversion in (u, v) and ordinary Fourier inversion in b . The small cube gives the Gaussian density after the change of variables $\zeta = \sqrt{n}(u, v, b)$. The covariance lower bound supplies domination. The remaining small annulus is integrable by part (ii) of Theorem 1.4; the compact/moderate and very-high ranges are integrable by parts (iii)–(iv). The Riesz projector surrounding the central insertion converges to the tilted stationary expectation. Integrating the last Gaussian coordinate over the requested unnormalized roof interval gives the exact conditional Gaussian interval probability. $\square$

Theorem 1.6 (Round-thirteen A2 closure). *The compact Sinai radius family has a uniform parent-branch anisotropic bundle, joint arithmetic/UNI control, an integrable estimate on the entire Fourier line, and the vector–count–roof density LLT of Theorem 1.5.*

Proof. Combine Theorems 1.1, 1.3, 1.4, and 1.5. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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